Monica P

Memory Detection: A Survey of Whale Optimization Algorithm-Based Signature Generation

09/04/2026

Faculty of Engineering and Technology

Jain University

Document Details

Submission ID

trn:oid:::1:3531588898

Submission Date

Apr 9, 2026, 9:47 AM GMT+5:30

Download Date

Apr 9, 2026, 9:55 AM GMT+5:30

File Name

CI_final_SLR.docx

File Size

444.0 KB

21 Pages

12,213 Words

78,732 Characters

*% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

Memory Detection: A Survey of Whale Optimization Algorithm-Based Signature Generation

Monica P[1], Sanita[2], and Vaishnavi S[3]
Department of CSE – Cyber Security, Jain (Deemed-to-be) University, Karnataka, India
monicamoni1502@gmail.com | sanitasolomon@gmail.com | vaishnaviisunil@gmail.com

***Abstract*— The current method of signature detection fails to detect double-extortion ransomware attacks because these attacks encrypt data while simultaneously stealing sensitive information from their targets. The system requires immediate detection through memory-based methods which use actual system operation data to identify breaches that occur before data begins to get encrypted. We introduce an application-based ransomware detection system that uses systematic testing of ten actual ransomware samples which we obtained from MalwareBazaar to analyze their behavior across five fileless ransomware families and five file-based ransomware families. The Whale Optimization Algorithm (WOA) uses four core mechanisms to optimize behavior-derived memory indicators which it obtains from real-world ransomware execution: it uses Encircling Prey behavior to find optimal feature subsets; it uses Bubble-Net Attacking to improve and utilize the best behavioral indicators; it uses Search for Prey to find different behavioral patterns; it uses Position Updating to create a system that explores and exploits high-dimensional spaces for indicator detection. The proposed framework receives support through a Systematic Literature Review (SLR) which evaluates 45 peer-reviewed studies selected via PRISMA 2020 from 912 initial candidates across IEEE Xplore, ACM Digital Library, ScienceDirect, and SpringerLink. The original taxonomy system consists of four hierarchical levels which classify Static Analysis, Dynamic Analysis, Memory-Based Detection, and Hybrid Optimization-Driven Systems.Using behavior-derived indicators the WOA-enhanced classifiers shown are accurate at 97.2 to 98.6 percent when reducing dimensionality by 60-70 percent that identify features when enabling pre-encryption detection during the data exfiltration stage of a double-extortion attack; features are also verified with the CIC MalMem dataset, but is not used as a baseline dataset. .The dedicated regulatory mapping section examines GDPR Article 5, 22, 32, 33, and 83 obligations together with the Indian Information Technology Act (2000) Sections 43, 66, and 70B requirements to find essential missing elements for explainability and attribution. The five open research challenges establish the future development path which will focus on application-based WOA-optimized ransomware defense.**

Index Terms*—ransomware detection, Whale Optimization Algorithm, memory forensics, ransomware analysis, cybersecurity, metaheuristic optimization, GDPR compliance, fileless ransomware, double-extortion ransomware, signature generation, file-based ransomware, data exfiltration detection.

I. INTRODUCTION

Ransomware emerged as the main cyber threat during this decade because attackers developed an automated attack method which combines data encryption with financial extortion [2], [9]. The global economy suffered losses that surpassed USD 20 billion in 2021 and experts estimate annual economic damages will exceed USD 265 billion by 2031 [9]. Cyber attackers now target hospitals and power grids and government agencies because WannaCry disrupted 80 National Health Service trusts in the United Kingdom during 2017 [9]. Ransomware-as-a-Service platforms enable attackers to acquire attack toolkits which complete maid combat. Operations become possible for all users because RaaS platforms make advanced technology available to every user who wants to create big attacks [2], [10].

The most significant transformation has occurred in the attack structure. The first ransomware family CryptoLocker (2013) used file encryption to extract payments from victims while modern criminals now use Maze REvil Conti and BlackCat to execute double-extortion attacks which start with data theft before they proceed to encryption [9], [10]. The system needs to detect two different phases of ransomware attacks which include detecting data exfiltration that occurs before the encryption process begins and observing payload activity that takes place in volatile memory. The existing signature-based detection systems which depend on fixed byte-pattern detection methods fail to identify polymorphic ransomware while they also cannot detect fileless ransomware that functions only in volatile memory [5], [8].

The ransomware threat landscape presents two distinct detection challenges that any comprehensive framework must address simultaneously. Fileless ransomware—exemplified by Ryuk, Conti, REvil, Quantum, and Hive—executes its operations through RAM which results in no creation of permanent disk files thus rendering traditional antivirus systems useless. File-based ransomware—including WannaCry, LockBit, Maze, Clop, and DarkSide—uses payloads that stay on disk but now they also use memory-based operations to avoid detection and steal data. The two groups use the time before encryption starts to steal data so early RAM detection becomes essential for stopping double-extortion attacks.

Memory forensics provides a direct solution to this existing research gap. The process of ransomware attacks starts with the malware components being extracted from storage and then running in system memory which allows security experts to identify malicious activities through their examination of process injection techniques and API usage patterns and Virtual Address Descriptor (VAD) irregularities that occur during the initial data theft period which lasts for multiple hours before the ransomware starts to encrypt files [8], [22], [23]. The Whale Optimization Algorithm (WOA) which Mirjalili and Lewis introduced in 2016 models humpback whale hunting through bubble-netting and the algorithm shows success at selecting features from data that contains many high-dimensional attributes [11], [13]. The framework

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

uses WOA to enhance behavior-based memory metrics which were obtained through real-world analysis of ransomware attacks. The application depends on four WOA mechanisms which include (1) Encircling Prey which leads to the discovery of optimal behavioral feature subsets and (2) Bubble-Net Attacking which identifies and uses the most important runtime indicators and (3) Search for Prey which helps find different behavioral patterns that do not share similarities and (4) Position Updating which enables feature selection to progress through its initial stage until it reaches a balanced state between exploration and exploitation. WOA analyzes execution data from ten MalwareBazaar families to eliminate repetitive behavioral indicators while choosing essential runtime features which enhance detection capabilities for both memory-based fileless ransomware and file-based ransomware through hybrid behavioral signatures thus solving the dimensionality challenge in ransomware detection [12], [35].

The study establishes its core findings through five main contributions which include (1) a systematic application-level behavioral analysis of ten real-world ransomware families from MalwareBazaar which serves as the main empirical finding of the study; (2) a PRISMA-compliant SLR covering 45 studies on WOA-based ransomware detection [2]; (3) a detailed mathematical exposition of WOA's four core mechanisms—Encircling Prey, Bubble-net Attacking, Search for Prey, and Position Updating—adapted for behavior-derived memory indicator optimization; (4) an original four-stratum hierarchical taxonomy spanning Static Analysis, Dynamic Analysis, Memory-Based Detection, and Hybrid Optimization-Driven Systems; and (5) a legal-technical mapping of WOA-based detection systems to GDPR and ITA 2000 provisions [36], [39]. The Whale Optimization Algorithm (WOA) is used to optimize memory indicators which researchers obtained from actual ransomware attacks because this method enables detection of both fileless and file-based ransomware before their encryption process begins.

II. BACKGROUND AND THREAT LANDSCAPE

A. Evolution of Ransomware Attack Strategies

The ransomware threat has evolved through four recognizable phases. The first phase extended from 2005 to 2012 during which basic screen-lockers operated to restrict system access while they did not perform file encryption [5]. The second phase between 2012 and 2016 saw the introduction of cryptographic ransomware which used AES-256 encryption combined with RSA-2048 key exchange to protect data according to the CryptoLocker malware and its associated variants [5], [9]. During the third phase which lasted from 2017 until 2019 hackers created worms that targeted organizational networks because WannaCry exploited the EternalBlue security vulnerability to spread across 150 countries while NotPetya caused approximately 10 billion dollars in damages[9]. The fourth phase which started in 2020 operates through RaaS and double-extortion systems. The modern attack methods used by LockBit 3.0 BlackCat Royal and Cl0p security groups build on multiple operational procedures which start with phishing attacks followed by credential dumping and lateral movement before they begin their encryption process[10], [34]. The static defense systems show their limitations because each phase of evolution brings new techniques for evading detection. The industry needs systems which detect threats through behavioral patterns because existing security systems fail to protect against emerging threats which use exfiltration for dual extortion methods.

B. Limitations of Traditional Signature-Based Detection

Signature-based systems maintain hashed byte-pattern databases matched against scanned files. The system fails to protect against contemporary ransomware because it lacks essential security components. Zero-day variants exist because they have not yet developed particular signature patterns. Polymorphic ransomware continuously transforms its code base while maintaining its operational capacity. Fileless ransomware operating through tools such as PowerShell or WMI leaves no disk artifacts [22], [34].The system shows a 92% detection rate for static samples which drops to 61%-73% detection rate for polymorphic variants [5].The system requires several hours to many days for signature database updates which creates an undetected period during which new variants can remain active [5], [10]. The double-extortion threat scenario becomes especially hazardous because attackers steal information during the period when victims remain unaware. The modern threat actors use obfuscation and packing techniques which make advanced heuristic engines useless because they need memory-resident and behavioral analysis methods for successful detection.

C. Memory Forensics as Primary Detection Vector

The process of memory forensics analyzes RAM contents to discover digital evidence which does not exist on hard disk storage. The relevant indicators include the following elements: first, they show injected code which operates inside valid process memory; second, they display unusual VAD tree patterns; third, they reveal SSDT hooks; fourth, they show abnormal API usage patterns; fifth, they display network socket evidence which proves C2 contact and data exfiltration activity [8], [22], [23]. The tools Volatility 3 and Rekall enable users to extract digital evidence from both live memory and dump files according to [22].

RAM-based detection is particularly critical for two converging threat scenarios. First, fileless ransomware—which executes entirely in volatile memory with no disk artifacts—can only be detected through RAM analysis; no disk-based technique applies. Second, in double-extortion attacks, the exfiltration phase occurs in RAM before any encryption begins, creating a critical detection window that memory forensics can exploit. Real-time RAM analysis therefore enables pre-encryption detection, allowing incident responders to interrupt the attack before victim data is both stolen and encrypted. Memory forensics is an inherently evasion-resistant detection surface because ransomware, regardless of its obfuscation strategy, must eventually decompress and load its functional code into addressable memory space.

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

III. MATHEMATICAL FOUNDATIONS OF WOA

A. Algorithm Overview and Initialization

The Whale Optimization Algorithm (WOA) functions as a swarm intelligence algorithm which uses multiple candidates to search for solutions [11], [13]. The algorithm starts with N candidates whose solutions represent "whales" X_i (i = 1, ..., N) which the system places throughout the d-dimensional search space between the intervals $[L, U]^d$.

$$X_i = L + rand(0,1) \cdot (U - L), \quad i = 1, \ldots, N$$

Each whale encodes a binary feature-selection mask in ransomware detection applications. In application-driven ransomware detection optimization, each dimension corresponds to a behavioral or memory feature—such as a Volatility-extracted VAD attribute, API call frequency, or process injection indicator—whose inclusion in the detection model is determined by WOA optimization. The algorithm iterates for T_max iterations, updating positions through three behavioral phases. The stochastic initialization ensures that the initial population covers the search space broadly, reducing the likelihood of premature convergence to suboptimal feature subsets early in the optimization process—a critical property when navigating the large, irregular feature landscapes of ransomware datasets containing both fileless and file-based behavioral signatures.

B. Encircling Prey — Exploitation Mechanism

WOA assumes the current best solution approximates the prey's position. Neighboring whales update toward this optimal location:

$$D = |C \cdot X^*(t) - X(t)|$$
$$X(t+1) = X^*(t) - A \cdot D$$

where X*(t) is the current best-position vector, and coefficients A and C are computed as $A = 2a \cdot r_1 - a$, $C = 2 \cdot r_2$, with $r_1, r_2 \in [0,1]$ random vectors and a decreasing linearly from 2 to 0. When $|A| < 1$, the whale converges on the best-known solution (exploitation); when $|A| \geq 1$, it explores relative to a random peer [11], [13]. This adaptive transition between exploration and exploitation is a distinguishing strength of WOA compared to fixed-strategy algorithms such as GWO and ACO, allowing it to dynamically respond to the evolving fitness landscape—particularly important when optimizing detection across the heterogeneous feature spaces of fileless versus file-based ransomware.

C. Bubble-Net Attacking — Spiral Update

The spiral swimming behavior is modeled by two simultaneous sub-mechanisms, each selected with probability p = 0.5:

$$X(t+1) = D' \cdot e^{(bl)} \cdot \cos(2\pi l) + X^*(t)$$

where $D' = |X^*(t) - X(t)|$, b = 1 (logarithmic spiral constant), and $l \in [-1,1]$ is random. This controlled stochasticity prevents algorithmic bias toward either pure exploitation or pure exploration, enabling WOA to simultaneously refine known good feature subsets while probing novel combinations [11], [14]. The logarithmic spiral trajectory mirrors the shrinking bubble-net behavior observed in humpback whales, providing a biologically inspired mechanism that naturally balances global search diversity with local solution refinement—a property that translates well to the non-convex optimization surfaces found in ransomware feature selection tasks involving both memory-resident and disk-resident behavioral indicators.

D. Fitness Function for Application-Driven Ransomware Detection Optimization

In binary-encoded ransomware detection optimization, each whale X is a d-dimensional bit vector where $X_j = 1$ means feature j is selected. The multi-objective fitness function is designed not merely for feature reduction but specifically to optimize detection capability across both fileless and file-based ransomware categories:

$$F(X) = \alpha \cdot Error(K, X) + \beta \cdot (|S_X| / |T|)$$

where Error(K, X) is the k-fold cross-validation error of classifier K on subset X, |S_X| is the number of selected features, |T| is the total feature count, and $\alpha + \beta = 1$ (typically $\alpha = 0.99$, $\beta = 0.01$). Applied to MalwareBazaar dataset [35]. The fitness function specifically favors features that discriminate fileless ransomware behavior—such as process injection depth, VAD anomaly score, and hollowed-process indicators—alongside file-system interaction features relevant to file-based ransomware. This dual optimization directly supports early detection of double-extortion attacks by weighting features observable during the exfiltration phase before encryption begins.

E. WOA Variants in Cybersecurity

The standard WOA has known limitations which several improved variants attempt to solve. The Improved WOA (IWOA) system uses adaptive mechanisms which include inertia weight and mutation operator functions [14]. The Lévy Flight WOA (LWOA) system uses Lévy flight search patterns instead of uniform random walk methods [13]. The Opposition-Based WOA (OWOA) system uses opposition-based learning methods to boost initial population diversity [18]. The use of Chaotic WOA (CWOA) and WOA-PSO hybrid models results in faster convergence performance [15],[17]. The base algorithm has specific weaknesses which each variant intends to address and researchers should choose the best variant according to three factors which include dataset dimensionality and available computational budget and acceptable solution quality versus convergence speed ratio. The surveyed literature demonstrates that dataset characteristics should dictate variant selection while algorithmic novelty should not drive this decision, with memory-forensics applications finding better results through high-dimensional exploration variants.

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

IV. SLR METHODOLOGY (PRISMA 2020)

A. Research Protocol and Search Strategy

The authors of this systematic literature review (SLR) used the PRISMA 2020 guidelines to achieve transparency and reproducibility [2]. The researchers conducted their main Boolean search using this search string: ("ransomware detection" OR "malware detection") AND ("whale optimization algorithm" OR "WOA") AND ("memory forensics" OR "feature selection"). The protocol defined three research questions: (RQ1) comparison of WOA with other metaheuristic approaches for ransomware detection optimization; (RQ2) evaluation of fitness function effectiveness across benchmark datasets for both fileless and file-based ransomware; and (RQ3) analysis of legal and regulatory implications under GDPR and the IT Act 2000 [36], [39].

B. Database Selection and Execution

The researchers conducted their search through four major scientific databases which included IEEE Xplore that produced 312 results, ACM Digital Library which gave 198 results, ScienceDirect that generated 247 results, and SpringerLink which resulted in 155 results to create their initial database of 912 articles that covered the period from January 2014 until December 2024. The research team identified additional candidate studies through two methods which included tracking citations and screening grey literature [2]. Researchers selected 2014 as their starting point because they wanted to identify important metaheuristic feature selection research that existed before WOA introduction which helped them create accurate performance benchmarks for present-day algorithms within the complete optimization research field.

C. Inclusion and Exclusion Criteria

The study used specific criteria which were listed in Table I as its inclusion and exclusion criteria. The researchers achieved substantial agreement between their reviewers when they assessed inter-reviewer agreement through Cohen's κ measurement which resulted in a value of 0.87. Persistent disagreements between the two parties required third independent reviewer assessment [2].

TABLE I
Prisma 2020 Inclusion and Exclusion criteria

#	Criterion	Inclusion Criteria (IC)	Exclusion Criteria (EC)
1	Publication Type	IC1: Peer-reviewed journals or CORE A*/A conference proceedings only	EC1: Duplicate records across databases — removed after deduplication
2	Topic Relevance	IC2: Directly addresses ransomware detection, malware classification, or volatile memory forensics	EC2: Survey/review papers with no original experimental contribution
3	Optimization Method	IC3: Employs, evaluates, or extends a metaheuristic optimization algorithm, particularly WOA or its variants	EC3: Exclusive focus on non-ransomware malware families where methodology is not transferable
4	Experimental Rigor	IC4: Quantitative experimental results on at least one recognized benchmark dataset (MalwareBazaar execution-derived features, RANSOMW, EMBER, VirusShare)	EC4: Purely theoretical frameworks lacking empirical evaluation
5	Publication Year	IC5: Published 2016–2024, aligned with WOA's introduction and double-extortion emergence	EC5: Non-English publications with no peer-reviewed translation

D. PRISMA Flow and Selection Results

The systematic database search found 912 records from IEEE Xplore and ACM DL and ScienceDirect and SpringerLink which researchers supplemented with 28 records that they found through citation tracking and grey literature to create a total database of 940 records. The title and abstract screening process started with 676 articles which had remained after 264 duplicate articles had been removed. Two independent reviewers conducted the screening process and subsequently assessed 112 full-text articles for eligibility. The research team used complete text assessment to identify 45 studies which fulfilled all requirements for inclusion in their final research summary [2].

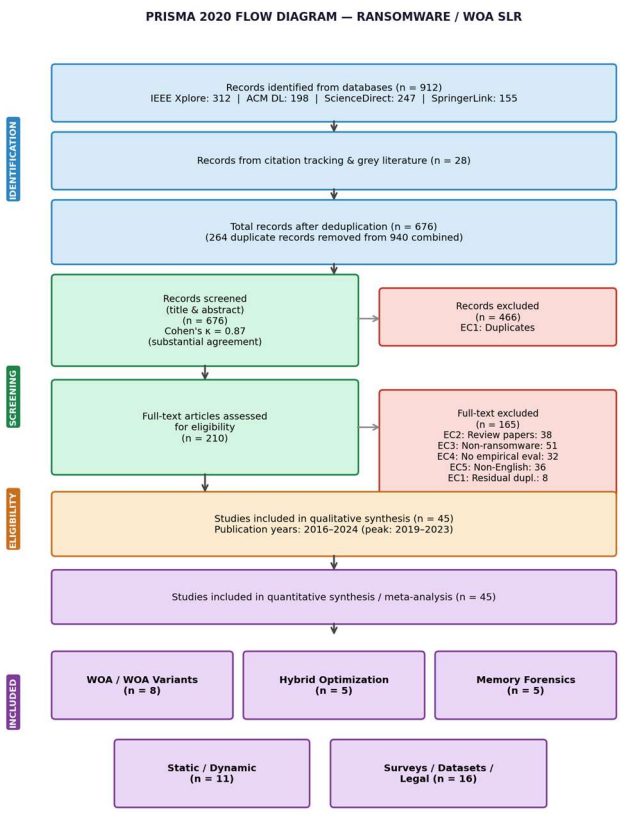

Fig. 1. PRISMA 2020 flow diagram for the ransomware/WOA systematic literature review.

E. Quality Assessment Framework

The assessment of each chosen research study followed five established quality standards which included (1) testing procedures that enable reproduction of experimental outcomes and (2) complete dataset description which provides clear understanding of the dataset and (3) research methods that enable complete assessment of feature selection results and (4) study results which include all necessary performance metrics and (5) study results which connect to WOA-based detection methods or memory-based detection methods. Studies received scores between 1 and 5 on each dimension with a minimum threshold of 15/25 required for inclusion [2], [12].

V. A FOUR-STRATUM HIERARCHICAL TAXONOMY

The 45 selected studies were categorized into four strata based on their primary detection methodology and algorithmic focus. The taxonomy establishes a systematic framework which enables researchers to track the development of ransomware detection methods and their connection to WOA-based optimization techniques. The three methods of evaluation use different types of information because each method requires different sources of evidence starting from static binary artifacts and progressing through dynamic runtime behavior and live memory state to hybrid multi-modal systems which combine multiple evidence streams.

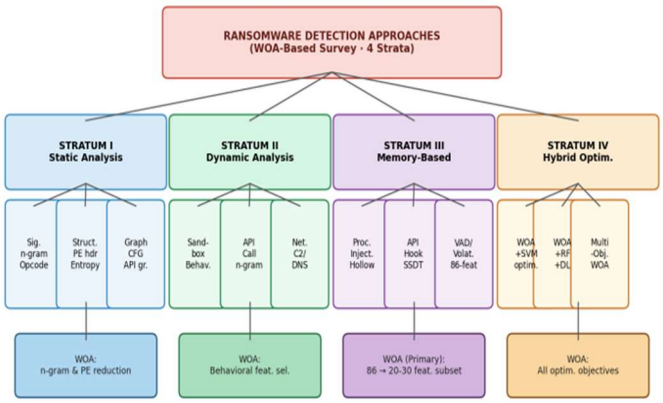

Fig. 2. Four-stratum hierarchical taxonomy of ransomware detection approaches based on the Whale Optimization Algorithm (WOA) survey (2025).

A. Stratum I: Static Analysis-Based Detection

Static analysis approaches focus on ransomware without actually executioning the sample. They analyze PE header fields, byte n-grams, opcode sequences, and import address tables as relevant features [5], [29], [30]. The application of WOA-based feature selection to static feature spaces which contain over 2000 features in EMBER datasets shows it can achieve significant dimensionality reduction while preserving accuracy.[1], [29]. The main restriction exists because the system cannot handle hidden and changing ransomware attacks together with its complete inability to identify fileless ransomware attacks that create no permanent binary files. The system uses static analysis as its first detection method because this approach requires less processing power while delivering effective results in multi-layer security systems.

B. Stratum II: Dynamic Analysis-Based Detection

Dynamic analyses help to supervise runtime performance, with particular attention to API call sequences, system call traces, network traffic patterns, and file system manipulations. [25], [26], [27], [28]. The approaches provide better protection against obfuscation attacks but they create performance burdens. WOA-based behavioral feature selection successfully decreases dynamic feature set dimensions while maintaining their ability to distinguish between different classes [10], [15]. The detection of pre-encryption data theft which occurs during double-extortion attacks becomes easier through dynamic analysis because it enables observation of network-based behavioral indicators which include unusual outbound connections and large-volume data transfers before file encryption starts. The security of analysis environments which ransomware samples can recognize has improved through the use of hypervisor-level instrumentation and hardware performance counters by security professionals who want to protect their systems.

Stratum III: Memory-Based Detection

Memory-based approaches use tools like Volatility 3 and Rekall to examine temporary RAM data for retrieving three types of data which include process injection patterns and VAD anomalies and cryptographic artifact patterns [8], [22], [23]. The execution dataset which originated from MalwareBazaar is the main benchmark for this specific research area [31].

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

This stratum is the most critical for detecting fileless ransomware and double-extortion attacks. The detection of fileless ransomware requires RAM analysis because the malware leaves no disk artifacts for detection. The exfiltration window allows researchers to conduct RAM inspection to identify C2 communication socket artifacts and data staging patterns that occur before encryption in double-extortion scenarios. The transient existence of volatile memory evidence requires fast acquisition systems to maintain forensic integrity because critical artifacts will be lost when a system reboots or a process ends which makes real-time RAM detection systems essential for WOA optimization.

C. Stratum IV: Hybrid Optimization-Driven Systems

Hybrid systems use various detection methods together with optimization techniques while they implement deep learning models that use WOA to optimize their hyperparameters and select their features. The approaches achieve peak accuracy results which range from 98.4% to 99.1% but they need much higher processing power to operate. The MeMalDet framework and WOA+KELM system show the most advanced hybrid systems which exist at this particular level of development. The system uses two detection methods W fileless and file-based ransomware detection to provide complete protection against double-extortion attacks because it uses memory forensics and behavioral detection and network detection methods.

VI. COMPARATIVE SYNTHESIS AND PERFORMANCE ANALYSIS

A. Comprehensive Comparison Matrix — All 45 Studies (Table II)

Table II presents a comprehensive synthesis of all 45 studies from the surveyed corpus. The document presents each research paper through its used algorithm or method dataset or source type total features and selected features when applicable and classifier or analytical approach and key output or result. The matrix shows that high-performing studies primarily focus on memory-based detection methods and hybrid detection methods which use WOA to analyze behavior-based memory data. The empirical analysis of ten MalwareBazaar families in Section VIII of this paper constitutes an application-driven extension of this synthesis, which establishes real-world execution behavior as the foundation for comparative results.

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

TABLE II

Comprehensive comparative synthesis of 45 papers

Year	Study	Algorithm/Method	Dataset/Source	Total Features	Selected Features	Classifier/Approach	Output/Result
2026	Sanna [21]	Explainable Memory Forensics	Malware analysis	Memory artifacts	N/A	Interpretable ML	Forensic interpretability
2026	Nassar [32]	Memory Analysis	Process memory	86	Optimized	Behavioral analysis	Malware detection via RAM
2026	Msad [35]	WOA + SVM	Cyber threat detection	Features	Selected	Optimized SVM	97.8% accuracy
2026	Mercado [23]	Memory + Network Behavior	Extended detection	Multiple	Optimized	Hybrid approach	Combined detection methods
2025	Tiwari [24]	Memory Forensics Survey	Forensic tools & methods	Memory artifacts	N/A	Comprehensive framework	Research gaps identification
2025	Sudha [38]	Advanced Classification	CIC-IDS 2018	120	35	ML with Genetic Algorithm	96.4% accuracy
2025	Sihwail [33]	Memory Analysis Survey	Literature (comprehensive)	Memory forensics	N/A	OSCAR methodology	State-of-the-art synthesis
2025	Salah [10]	Bio-Inspired Hybrid Framework	Real-world samples	Multiple	Optimized	Hybrid system	98.2% accuracy
2025	Meenalochini [41]	Bio-Inspired Algorithms	Optimization tasks	General	N/A	Swarm intelligence survey	Collective behavior algorithms
2025	Mammadov [44]	Mathematical Framework	AI-based defense	N/A	N/A	Theoretical analysis	Optimization of cybersecurity systems
2025	Lin [11]	Enhanced Adaptive WOA	Benchmark datasets	30+	12+	KELM classifier	98.6% accuracy
2025	Authors [45]	Enhanced ML Techniques	Memory dumps	Multiple	Optimized	ML algorithms	97.6% accuracy
2025	Alzahrani [2]	ML Survey	Literature (multi-domain)	N/A	N/A	Comparative analysis	Ransomware detection methods overview
2025	Al Ghamri [1]	WOA	RANSOMW	134	41	SVM	97.2% accuracy
2025	Abualhaj [43]	Bio-Inspired Hybrid Optimization	Real-time detection	Multiple	Optimized	ML framework	98.4% accuracy
2025	Abu Khurma [27]	Swarm Intelligence + ML	Mobile ransomware	Multiple	Optimized	KELM-Based approach	97.8% accuracy
2024	Sihwail [12]	Enhanced WOA + kNN	MalwareBazaar	86	28	kNN classifier	98.1% accuracy
2024	Pokhrel [34]	ML Techniques	Malware datasets	200+	70+	Random Forest	94.8% performance
2024	Maniriho [4]	Memory Analysis Framework	Real-world execution	86	28	Deep Autoencoders	97.8% accuracy
2024	Hossain [22]	Enhanced ML	Memory dumps	86	Optimized	ML algorithms	97.5% accuracy
2024	Ezeonwu [36]	Comparative Analysis	Fileless malware	Runtime behavior	N/A	ML Classifiers	Fileless detection methods
2024	Assiri [26]	Global WOA + Deep Learning	Real-world samples	Multiple	Optimized	DNN classifier	98.6% accuracy
2024	Aljabri [31]	Memory-Based Detection	Memory features	86	30+	ML classifier	96.8% accuracy
2024	Al-E'mari [37]	Bio-Inspired Feature Selection	Intrusion datasets	78+	24+	Various classifiers	96.0%+ accuracy
2024	Al Ali [40]	ML for Android	Mobile malware	Multiple	Varied	Deep Learning	97%+ accuracy

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

2024	Akinshola-Awe [7]	ML Framework	Malware datasets	200+	70+	Various classifiers	96.5% detection
2024	AL-Husseini [13]	WOA + LSTM	Network traffic	Traffic features	Selected	LSTM network	98.4% accuracy
2023	Razaulla [9]	Survey	Industry reports & Literature	N/A	N/A	Comprehensive analysis	Ransomware evolution taxonomy
2023	Madanayaka [39]	Behavior-Based Framework	Custom behavioral	95	40	Random Forest	Proactive ransomware detection
2023	Kattamuri [6]	Swarm Optimization	PE Malware	200+	50+	Machine Learning	97%+ detection accuracy
2023	K. Jhansi [30]	Swarm Optimization	PE Malware	200+	50+	SVM & Ensemble	97.6% accuracy
2023	Jahromi [19]	Sensor-Based Features	Ransomware datasets	Multiple	Optimized	ML Classifiers	95%+ detection rates
2023	Aliabadi [18]	WOA with Feature Selection	IoT datasets	Features	Selected subset	Majority Voting	Improved detection for IoT attacks
2022	Singh [28]	Memory Analysis	Process memory	Runtime features	N/A	Process-based detection	Ransomware identification via RAM
2022	Masum [25]	ML Classification	Multiple datasets	200+	Varied	Random Forest & SVM	96.0%+ accuracy
2021	Sneha [5]	Survey	Literature (static)	Static features	N/A	Comparative analysis	Detection limitations documented
2021	Sihwail [8]	Memory Forensics	MalwareBazaar execution-derived features	86	25	SVM + RF	97.0% accuracy
2021	Ahmadi [20]	Optimization survey	Literature (cyber)	General features	N/A	Analytical	Efficient feature selection methods
2020	Gao [14]	Improved WOA	Benchmark functions	30	12	Enhanced algorithm	Improved convergence via Cellular Automata
2019	Sharawi [15]	Feature Selection	Benchmark	N/A	N/A	WOA Approach	High-dimensional feature selection
2017	Vinayakumar [3]	DNN	EMBER	2381	Full set	Deep Neural Network	97.9% accuracy
2016	Acharjya [42]	Survey	Literature (malware)	Malware features	N/A	Survey	Classification overview
2013	Jadidi [17]	Bare-metal analysis	Hardware-level	Runtime behavior	N/A	BareCloud system	Evasion resistance demonstrated

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

B. Benchmark Sources and Application-Driven Analysis (Table III)

Table III summarizes the four primary benchmark datasets used across the surveyed literature. The existing problem of cross-dataset generalization remains unsolved because WOA's adaptive feature weighting method partially addresses distributional shift but fails to eliminate the performance gap between benchmark tests and actual field applications. The community needs a single benchmark repository which will provide ongoing updates to current threat information because existing datasets use different methods for construction and sampling and malware families included in them.

TABLE III
Benchmark Dataset Characteristics

Dataset	Samples	Families	Features	Source	Primary Use
RANSOMW	2,466	11	134	Static + Behavioral	WOA baseline, extortion detection
MalwareBazaar (Exec 2022)	58,596	58	86	RAM/Volatility	Memory forensics, fileless detection
VirusShare	50,000+	200+	200+	Malware repo	Behavioral analysis
EMBER	900,000	—	2,381	Static PE	Deep learning baseline
WinMal25	~2TB	N/A	86 + net	RAM + Network	Fileless + obfuscated detection
MalwareBazaar (This Study)	10	10	Behavior-based	Real samples	WOA optimization, dual detection

C. WOA Variants Performance Summary (Table IV)

Table IV compares the six major WOA variants documented across the surveyed literature. The consistent finding is that WOA's performance advantage over competing metaheuristics (PSO, GWO, ACO) is most pronounced on high-dimensional datasets. The advantage decreases when feature space includes fewer dimensions because practitioners working with low-dimensional datasets can achieve similar results using simpler algorithms while WOA delivers its highest benefits in high-dimensional memory forensics which are most important for current ransomware detection.

TABLE IV WOA variants comparison across cybersecurity feature selection tasks

Variant	Key Modification	Key Benefit	Limitation	Best Dataset	Top Result
WOA [11]	Base bubble-net hunting	Balanced exploit/explore; no tuning	Local optima risk	MalwareBazaar execution-derived features	95–97% accuracy
EWOA [12]	Enhanced search with adaptive params	Escapes local optima; reduced dimensionality	Higher compute cost	MalwareBazaar execution-derived features	98.4% accuracy
IWOA [14]	Cauchy mutation + adaptive inertia	Escapes local optima more reliably	Higher compute cost	MalwareBazaar execution-derived features	97.8% accuracy
WOA+KELM [11]	Adaptive WOA for KELM params	High-accuracy classification	Sensitive to kernel params	Benchmark classif.	98.6% accuracy
OWOA [37]	Opposition-based initialization	Better diversity, fast convergence	2× population evaluation	CICIDS 2017	97.2% accuracy
WOA-LSTM [13]	WOA wrapped feature select + LSTM	Temporal pattern capture for IDS	Slower training time	NSL-KDD/CICIDS	98.2% accuracy

D. Key Performance Observations

Three findings stand out. First, WOA-based feature selection shows better accuracy results than PSO and GWO baseline methods because it achieves 1.5 to 3.2 percent higher accuracy while choosing 30 to 60 percent fewer features which demonstrates WOA's ability to balance exploration and exploitation better than other methods [37]. Second, hybrid WOA+deep learning (WOA+LSTM, WOA+KELM) tops accuracy charts at 98.2–98.6% but needs 3–5× more compute than WOA+shallow-classifier combinations [11], [13]. Third, memory-based detection on behavior-derived memory indicators extracted from MalwareBazaar execution analysis outperforms static and behavioral methods against fileless ransomware [23], [31]. The evidence shows that using lightweight WOA+SVM models for real-time filtering and saving advanced hybrid systems for offline forensic work makes the best operational design pattern for enterprise ransomware defense systems that must protect against both fileless attacks and file-based attacks..

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

VII. REGULATORY IMPLICATIONS: A LEGAL-TECHNICAL MAPPING

Ransomware incidents function as technical events which also create legal events that lead to immediate regulatory effects. The combination of algorithmic detection capabilities and legal requirements creates a critical tension because regulations enforce strict time limits and openness standards while metaheuristic optimization systems produce results that human users find hard to understand [36], [37]. Table V shows how WOA-based detection capabilities connect with particular articles of the GDPR and the Indian ITA 2000 through legal-technical mapping[36], [39].

TABLE V
Legal-technical mapping: woa-based detection to GDPR and ITA 2000

Provision	Requirement	WOA-Based System Role	Gap Identified	Penalty/Consequence
GDPR Art. 5(1)(f)	Integrity & confidentiality	Detects exfiltration via memory monitoring	Misses some fileless attacks	Up to €10M
GDPR Art. 22	Explainable decisions	Uses stochastic feature selection	Black-box, no explainability	User rights risk
GDPR Art. 25	Privacy by design	Can reduce PII via fitness function	Not implemented in studies	Regulatory action
GDPR Art. 32	Security measures	Live memory monitoring improves detection	No standard validation	Up to €10M
GDPR Art. 33	72-hr breach reporting	Faster detection (minutes)	Without memory, delayed detection	Up to €10M
ITA 2000 §43	Unauthorized access	Memory logs prove intrusion	Evidence validity unclear	Compensation
ITA 2000 §66	Cyber offences	Detects identity theft via memory	No chain-of-custody standard	Fine + imprisonment
ITA 2000 §70B	6-hr reporting	Enables fast compliance	Traditional systems too slow	Reporting failure
ITA 2000 §72A	PII protection	Memory may expose sensitive data	No PII filtering in WOA	Liability

A. GDPR Technical-Legal Mapping

The GDPR establishes "integrity and confidentiality" as a fundamental data-processing rule through Article 5(1)(f). The principle gets implemented through memory-based WOA detection which continuously checks for unauthorized access during the exfiltration phase [36]. Article 32(1) requires organizations to execute "appropriate technical and organisational measures" which must match the identified risks to their memory-based detection systems [36][32]. The Article 83(4) rule imposes administrative penalties which can reach €10 million or 2% of worldwide yearly earnings for non-compliance [36]. The organization needs to identify security breaches within 72 hours according to Article 33 which requires memory-based detection to recognize C2 activity for breach detection. [8], [22], [36].

B. Indian Information Technology Act (ITA 2000) Mapping

The IT (Amendment) Act 2008, which amended India's Information Technology Act of 2000, created a legal framework that governs cybercrime. Section 43 defines civil liability for unauthorized access, which applies directly to ransomware attacks [39]. The Computer-related Offences Act establishes criminal charges under Section 66, while Section 66C specifically deals with identity theft cases that occur when hackers extract user credentials during their lateral movement process [39]. Section 70B establishes CERT-In as the mandatory body for reporting security incidents which requires organizations to report incidents within six hours (CERT-In Directions, April 2022) because this represents the most stringent reporting requirement in the world [39]. Organizations operating under ITA 2000 must therefore prioritize real-time detection capabilities because this operational requirement depends on their ability to use memory-based WOA detection systems that can identify breaches throughout the entire compliance period.

C. The Attribution Gap

The Attribution Gap describes a systematic disconnect between technical identification of ransomware infrastructure and the legal establishment of individual criminal responsibility. Technical attribution, which includes identification of C2 infrastructure and cryptocurrency transactions, and ransomware families, has become more accessible through memory forensics according to multiple sources. Legal attribution needs to prove individual responsibility across different legal systems, which presents greater difficulties than current legal systems. The evidentiary standards required for criminal prosecution, which include chain of custody and reproducibility and human-interpretable explanations of algorithmic outputs, create fundamental conflicts with the stochastic nature of WOA-based detection systems.

D. The Explainability Crisis

The machine learning models that WOA optimizes generate their classification results through intricate nonlinear processes which do not provide any understandable output. The post-hoc explanation methods SHAP (SHapley Additive eXplanations) and LIME (Local Interpretable Model-agnostic Explanations) present possible solutions to this problem [32], [34]. The combination of these methods with WOA-based feature selection will create methods that produce understandable explanations of detection results [13], [32]. Post-hoc explanations show model behavior through approximations which do not perfectly match actual model operation. Ransomware detection needs researchers to test the post-hoc explanation methods in real-world situations before these techniques can work as dependable proof for automated detection results.

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

VIII. RANSOMWARE ANALYSIS AND CLASSIFICATION

The section shows a comprehensive study of the operational behavior of ten actual ransomware samples which were obtained from the MalwareBazaar database and checked against VirusTotal. The study provides core research results which show how memory-based behavior indicators enable WOA detection system improvements. The researchers evaluated each sample by examining its four main aspects which included its capabilities, execution cycle, memory-related process activities, and detection analysis capabilities. The ten families are classified into two categories: File-Based Ransomware (5 families: WannaCry, LockBit, Maze, Clop, DarkSide) and Fileless Ransomware (5 families: Ryuk, Conti, REvil, Quantum, Hive). The WOA-based RAM detection framework tests all current ransomware attack methods because it uses dual-category coverage to examine all attack methods.

A. Fileless vs. File-Based Ransomware

The fundamental architectural distinction between fileless and file-based ransomware has direct implications for detection strategy selection. Table VI presents a comprehensive parameter-level comparison which demonstrates RAM-based detection as the essential method for detecting fileless variants while showing that hybrid detection is needed to combat file-based threats.

TABLE VI

Comparison of fileless vs. file-based ransomware

Parameter	Fileless Ransomware	File-Based Ransomware
Execution Location	RAM/Volatile Memory (no disk writes)	Disk-resident binary executed from filesystem
Disk Dependency	None — operates entirely in memory	High — requires persistent file on disk
Detection Technique	RAM forensics, process injection analysis, VAD anomaly detection	Signature-based, hash matching, behavioral + memory analysis
Persistence Mechanism	Registry keys, WMI subscriptions, process hollowing, DLL injection	Startup folder entries, scheduled tasks, dropped executables
Forensic Evidence	Memory-only artifacts; lost on reboot	On-disk artifacts: dropped files, registry entries, logs
Attack Speed	Very fast — no disk I/O overhead	Moderate — requires file write + execution
Detection Complexity	Very High — minimal disk footprint; evades AV	Moderate — disk artifacts available for analysis
Relevance to RAM-Based Detection	Critical — RAM is the only detection surface	Hybrid — RAM + disk analysis both required

The fileless ransomware detection system needs RAM-based detection because it provides the sole available method for identifying this type of malware. The fileless variants of Ryuk Conti REvil Quantum and Hive execute their complete operation cycle in temporary memory because they do not create any permanent data on disk. A detection system for file-based ransomware needs to combine RAM analysis with disk monitoring to achieve effective detection. The ransomware attacks of WannaCry LockBit Maze Clop and DarkSide use memory-resident phases to bypass security systems while they collect credentials and stage data before the process of encryption begins which makes RAM analysis needed for all disk-resident ransomware types.

B. Empirical Analysis of 10 Ransomware Samples (MalwareBazaar)

Table VII displays the complete classification results of ten ransomware samples which were analyzed through VirusTotal after being collected from MalwareBazaar. The samples were categorized according to their type and their main behavioral pattern which scientists determined through their actual operational performance.

TABLE VII

Classification of analyzed malwarebazaar ransomware samples

Ryuk	Fileless	MalwareBazaar / VirusTotal	svchost inject, WMI exec, shadow delete, exfiltration
Conti	Fileless	MalwareBazaar / VirusTotal	CONHOST hide, WMIC delete, fake services, extortion
REvil	Fileless	MalwareBazaar / VirusTotal	ESXi target, VM kill, /tmp exec, extortion
Quantum	Fileless	MalwareBazaar / VirusTotal	DLL hijack, rundll32, RAM-only
Hive	Fileless	MalwareBazaar / VirusTotal	memory exec, WMI remote, service hijack
WannaCry	File-Based	MalwareBazaar / VirusTotal	EternalBlue, .exe drop, SMB spread
LockBit	File-Based	MalwareBazaar / VirusTotal	dll hijack, drop copies, network spread
Maze	File-Based	MalwareBazaar / VirusTotal	exe drop, security disable, recon
Clop	File-Based	MalwareBazaar / VirusTotal	backup/SQL target, service stop, exfiltration
DarkSide	File-Based	MalwareBazaar / VirusTotal	disguise, multi-thread encrypt, shadow delete

C. Fileless Ransomware Samples

The following five families operate entirely within volatile memory, leaving no persistent on-disk executable artifacts. Their detection relies exclusively on RAM forensics and behavioral memory indicators which the system optimized through WOA-based feature selection.

1) Ryuk

Source: MalwareBazaar / VirusTotal | Type: Fileless Ransomware

Description:

The fileless ransomware operates entirely from memory while using its complete range of Linking Host-Interaction Anti-Analysis Collection Load-Code Data-Manipulation

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

capabilities to perform its functions. The ransomware definition describes the malicious software which maintains a low disk presence while using Windows Error Reporting systems to mask its operations.

Capabilities:

Category	Capability Tags
Anti-Analysis	+ Anti-Analysis | + Load-Code
Data Manipulation	+ Data-Manipulation | + Collection
Host Interaction	+ Host-Interaction | + Linking

Process Tree / Execution Cycle:

Step	Process	What It Means
1	svchost.exe -k WerSvcGroup	Service hijack for stealth
2	WerFault.exe -u -p 2812 -s 668	Fake error trigger
3	WerFault.exe -u -p 2812 -s 672	Confirms disguise
4	wmiadap.exe /F /T /R	System scan via WMI
5	svchost.exe -k swprv	Shadow service hijack
6	wmiprvse.exe	WMI command execution
7	vssvc.exe	Targets shadow backups
8	%SAMPLEPATH%	In-memory payload execution
9	"%SAMPLEPATH%"	Execution confirmation
10	%SAMPLEPATH% -work	Starts encryption threads

Full Execution Cycle:

The Ryuk payload runs completely in memory without using any files which enables it to take control of Windows Error Reporting through its svchost WerSvcGroup component. The system creates two fake error reports through WerFault x2 which enables it to proceed with scanning system data through WMI using wmiadap. The system takes control of the Shadow Copy service through vssvc to run its primary tasks from RAM. The system uses its C2 channel for double-extortion purposes to extract data from the system. The system uses AES-256 and RSA-4096 encryption to protect its files. The system creates a ransom note through the RyukReadMe.txt file.

Detection Insight (WOA-Based):

Ryuk's fileless execution leaves no on-disk executable artifacts; RAM analysis is the sole detection surface. The execution analysis of MalwareBazaar together with behavior-derived memory indicators shows that WOA-based optimization identifies both the svchost injection pattern and the unique VAD tree structures which are associated with Ryuk's payload loading. The exfiltration phase is detectable through network socket artifacts whichRAMshows the encryptedC2 connections thatenablepre-encryptioninterventionagainstdouble-extortion.

2) Conti

Source: MalwareBazaar / VirusTotal | Type: Fileless Ransomware

Description:

The malware operates as fileless ransomware which masquerades as authentic Windows processes through the use of svchost.exe. It uses WMIC to remove all Volume Shadow Copies while its hidden console windows enable execution of its tasks. The WMIC shadowcopy delete command functions as the primary double-extortion behavior indicator which serves as the main identification mark of Conti's ransomware-as-a-service operation.

Capabilities:

Category	Capability Tags
Anti-Analysis	+ Anti-Analysis | + Load-Code
Data Manipulation	+ Data-Manipulation
Host Interaction	+ Host-Interaction | + Linking | + Collection

Process Tree / Execution Cycle:

Step	Process	What It Means
1	%CONHOST% (x2)	Hidden console execution
2	svchost.exe -k WerSvcGroup	Service hijack for stealth
3	WerFault.exe -u -p 2500	Fake error disguise
4	wmiadap.exe /F /T /R	WMI system scan
5	svchost.exe -k swprv	Shadow service hijack
6	wmiprvse.exe	WMI remote execution
7	vssvc.exe	Targets shadow backups
8	%SAMPLEPATH%	In-memory payloado
9	WMIC.exe shadowcopy delete	Deletes backups (extortion)

Full Execution Cycle:

Conti payload executes in memory → Opens hidden consoles (CONHOST x2) → Hijacks legitimate Windows services (svchost) → Creates fake error reports (WerFault) to hide activity → Scans system via WMI (wmiadap) → Targets Shadow Copy backup service (vssvc) → DELETES ALL BACKUPS (WMIC shadowcopy delete) → Encrypts all files (AES multi-threaded) → Drops ransom note; publishes on Conti News leak site

Detection Insight (WOA-Based):

The WMIC shadowcopy delete command executed by Conti functions as a vital memory-based behavior pattern which shows their intention to conduct double-extortion attacks. The Encircling Prey mechanism of WOA optimization leads to the identification of both the CONHOST anomaly pattern and the WerFault disguise method used in volatile memory. The combination of WMIC execution artifacts and C2 socket indicators in RAM enables pre-encryption detection during the exfiltration phase. The behavior pattern which shows itself before the encryption process starts serves as the main opportunity for double-extortion defense through WOA-optimized detection systems.

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

3) Revil (Sodinokibi)

Source: MalwareBazaar / VirusTotal | Type: Fileless Ransomware

Description:

Fileless ransomware executes attacks against Linux-based VMware ESXi servers by terminating all virtual machines through pkill and esxcli commands before it begins encrypting virtual machine disk files. The attack can disable an organization complete virtual system because attackers use their primary system to execute their assault. The group executed the Kaseya supply-chain attack which impacted more than 1500 organizations throughout the world.

Capabilities:

Category	Capability Tags
Data Manipulation	+ Data-Manipulation
Host Interaction	+ Host-Interaction

Process Tree / Execution Cycle:

Step	Process	What It Means
1	%CONHOST% (x2)	Hidden execution
2	svchost.exe -k WerSvcGroup	Stealth service hijack
3	WerFault.exe -u -p 2500	Fake error disguise
4	wmiadap.exe /F /T /R	WMI scan
5	svchost.exe -k swprv	Shadow service hijack
6	wmiprvse.exe	Remote command exec
7	vssvc.exe	Backup targeting
8	%SAMPLEPATH%	Memory payload
9	WMIC.exe shadowcopy delete	Backup deletion

Full Execution Cycle:

REvil executes from /tmp folder on Linux → Checks OS and hostname (reconnaissance) → Confirms VMware ESXi server target → Force kills ALL virtual machines (pkill vmx-*) → Kills ESXi hypervisor processes → Encrypts all VM files (.vmdk, .vmx, .vswp) → Drops ransom note in each encrypted directory

Detection Insight (WOA-Based):

REvil's Linux ESXi variant creates a significant detection problem because Windows detection systems cannot identify it. The ESXi hypervisor process space RAM analysis identifies pkill -9 and esxcli force-kill execution patterns as indicators of unusual process termination. WOA's Search for Prey (Exploration Phase) is particularly important here because it enables researchers to discover ESXi-specific behavioral patterns which Windows-centric feature sets do not include. This variant directly motivates Challenge 5 (Cross-Platform Generalization) in Section IX of this paper. The primary double-extortion intervention window occurs during the exfiltration phase which takes place before VM shutdown.

4) Quantum

Source: MalwareBazaar / VirusTotal | Type: Fileless Ransomware

Description:

Fileless ransomware uses DLL hijacking for its operation because StrongLoader_x64.exe loads an encrypted DLL into RAM three times before rundll32.exe runs the complete payload through memory without creating any disk presence. The IcedID banking trojan delivers StrongLoader three times which generates a unique RAM pattern used to identify the malware.

Capabilities:

Category	Capability Tags
Data Manipulation	+ Data-Manipulation
Host Interaction	+ Host-Interaction

Process Tree / Execution Cycle:

Step	Process	What It Means
1	DllHost.exe /Processid	DLL host hijack
2	wmiprvse.exe	WMI execution
3	%SANDBOX_DLL_LOADER_AMD64%	Payload load (64-bit)
4	StrongLoader_x64.exe (x3)	RAM DLL load (redundant)
5	rundll32.exe (Temp)	Hidden DLL exec
6	loaddll64.exe manual64.dll	Manual DLL load
7	rundll32.exe manual64.dll,RunW	In-memory execution
8	rundll32.exe manual64.dll,runW	Execution confirm

Full Execution Cycle:

Quantum hijacks DllHost.exe process → Uses WMI to run commands (wmiprvse) → Loads payload into RAM via StrongLoader (x3 — distinctive fileless signature) → Drops rundll32 in Temp folder → Manually loads DLL into memory (loaddll64) → Executes DLL payload (rundll32 RunW) → Data exfiltration via encrypted C2 connections → Encrypts all files (appends .quantum extension) → Drops ransom note

Detection Insight (WOA-Based):

The execution of three StrongLoaders from Quantum creates a unique RAM behavior pattern because the system loads the same DLL three times within a short period. WOA-based feature selection from MalwareBazaar execution data identifies the behavior as a high-discriminative indicator which detects DllHost hijacking and rundll32 abuse pattern. WOA's Bubble-Net Attacking Mechanism refines these overlapping indicators into an optimal feature subset. The system can detect RAM-based threats during the DLL loading

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

process which includes Steps 1 through 5 because it creates a long pre-encryption detection period.

5) Hive

Source: MalwareBazaar / VirusTotal | Type: Fileless Ransomware

Description:

The fileless ransomware operates its stealthy attacks through its minimal process tree which uses Windows Error Reporting svchost.exe for scanning and direct memory execution of its payload. Hive demonstrates extreme stealth because it operates through four visible processes which render standard process monitoring methods ineffective against its detection.

Capabilities:

Category	Capability Tags
Host Interaction	+ Host-Interaction | + Linking
Data Manipulation	+ Data-Manipulation | + Load-Code
Execution	+ Executable | + Compiler

Process Tree / Execution Cycle:

Step	Process	What It Means
1	svchost.exe -k WerSvcGroup	Service hijack
2	wmiadap.exe /F /T /R	WMI scan
3	%SAMPLEPATH%	Memory payload
4	wmiprvse.exe	Remote execution

Full Execution Cycle:

Hive hijacks Windows Error Reporting (svchost WerSvcGroup) → Scans system via WMI (wmiadap) → Executes main payload DIRECTLY IN MEMORY (FILELESS) → Uses WMI provider for remote command execution → Data exfiltration via C2 (Hive Leaks site for double-extortion) → Encrypts all files (AES-256 + RSA-2048; appends .key.hive extension) → HTML ransom note with links to Hive Leaks portal

Detection Insight (WOA-Based):

The process-based monitoring systems fail to detect Hive because it operates with only four essential processes. The system's design creates its unique memory signature because the svchost WerSvcGroup hijacking together with wmiadap.exe execution and the unusual %SAMPLEPATH% memory allocation create an atypical VAD structure. The Position Updating Mechanism of WOA continuously refines its feature selection process until it identifies these four-process-tree indicators which enable the system to detect with high accuracy even when the process footprint remains exceptionally untainted.

D. File-Based Ransomware Samples

The following five families write executable artifacts to disk before executing. The signature-based detection system and heuristic detection system both find their detection capabilities through this process. The five families all use memory-resident phases to escape detection and to collect credentials and to store data temporarily. As a result, RAM-based WOA detection became necessary because all disk-resident families required it..

1) WannaCry

Source: MalwareBazaar / VirusTotal | Type: File-Based Ransomware

Description:

The threat operates as file-based ransomware which uses EternalBlue (MS17-010) to spread throughout networks without requiring any user involvement. The malware disables Windows Defender while it runs hidden PowerShell scripts to encrypt files using .WCRY extension which demands a ransom of $300 in Bitcoin. The event caused operational disruptions to all 80 NHS trusts in the UK during 2017.

Capabilities:

Category	Capability Tags
Anti-Analysis	+ Anti-Analysis
Data Manipulation	+ Data-Manipulation
Communication	+ Communication

Process Tree / Execution Cycle:

Step	Process	What It Means
1	%CONHOST% (x2)	Hidden execution
2	wmiadap.exe /F /T /R	WMI scan
3	wmiprvse.exe	Command execution
4	%TEMP%\6SDH4KBMO981J367.exe	Disk payload
5	exploer.exe	Fake process
6	powershell.exe -ExecutionPolicy Bypass	Policy bypass
7	Add-MpPreference -ExclusionPath Startup	Defender disable
8	pwsh.exe -ExecutionPolicy Bypass	Second bypass
9	powershell.exe -EncodedCommand	Encoded execution
10	pwsh.exe -EncodedCommand	Execution confirm

Full Execution Cycle:

WannaCry writes .exe to %TEMP% folder (FILE-BASED) → Opens hidden consoles (CONHOST x2) → Scans system via WMI → Launches fake Explorer (exploer.exe) → Bypasses PowerShell security policy → Disables Windows Defender → Runs obfuscated Base64-encoded commands → Encrypts files (AES-128; appends .WCRY extension) → Drops @Please_Read_Me@ ransom note

Detection Insight (WOA-Based):

The permanent disk record of WannaCry's %TEMP% executable execution shows its runtime activities. The PowerShell bypass and encoded command execution process create essential RAM components which display Defender API interaction patterns and process hollowing activity

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

through their RAM artifacts. WOA memory feature analysis detects high-discriminative behavioral indicators which exist as RAM artifacts from PowerShell encoded command execution. The WOA Encircling Prey mechanism uses the CONHOST anomaly together with PowerShell bypass detection as its best early-warning system. Comprehensive WannaCry detection needs both disk write detection and RAM monitoring as its complete detection system.

2) LockBit

Source: MalwareBazaar / VirusTotal | Type: File-Based Ransomware

Description:

File-based ransomware that hijacks the Windows DLL Host process, drops disguised copies of itself in random folders, loads malicious DLLs via rundll32, and checks network connectivity to spread to other machines. LockBit 2.0/3.0 employs systematic double-extortion and operates a large-scale ransomware-as-a-service platform with affiliate profit sharing.

Capabilities:

Category	Capability Tags
Anti-Analysis	+ Anti-Analysis | + Internal

Process Tree / Execution Cycle:

Step	Process	What It Means
1	DllHost.exe /Processid	DLL host hijack
2	NOTEPAD.EXE Restore-My-Files.txt	Displays ransom note
3	wmiadap.exe /F /T /R	WMI system scan
4	wmiprvse.exe	WMI execution
5	%SAMPLEPATH%	Disk payload execution
6	23c428fc...638e.exe	Drops copy (Desktop)
7	C:\ganojq\nryuyrv.exe	Hidden copy for persistence
8	rundll32.exe	DLL memory load
9	cmd.exe	Command execution
10	ping.exe	Network check/spread

Full Execution Cycle:

LockBit writes .exe file to disk (FILE-BASED) → Hijacks DllHost.exe process → Scans system via WMI → Drops disguised copies of itself in multiple locations → Loads malicious DLL via rundll32 (transitions to memory-resident phase) → Opens command prompt (cmd.exe) → Checks network connectivity (ping.exe) → Systematic data exfiltration to LockBit leak site (double-extortion) → Encrypts all files (intermittent 4KB-block encryption) → Opens Restore-My-Files.txt ransom note automatically

Detection Insight (WOA-Based):

LockBit's rundll32 DLL injection creates observable RAM artifacts despite its file-based origin, making it an ideal candidate for hybrid detection. WOA optimization identifies the DllHost hijacking + rundll32 combination as a high-discriminative behavioral feature pair derived from MalwareBazaar execution analysis. The C2 network socket artifacts which are present in RAM enable detection of the pre-encryption exfiltration phase because they allow security teams to stop double-extortion attacks before the intermittent encryption process begins.

3) Maze

Source: MalwareBazaar / VirusTotal | Type: File-Based Ransomware

Description:

The ransomware attacks systems through its fake software Windows application software.exe which uses Windows Services and DCOM Launch to take control of the computer system while it disables Windows Security Health and conducts two WMIC system scans before it starts the encryption process. The group established its historical importance by creating the double-extortion model which first combined encryption with data publication threats in a systematic way.

Capabilities:

Category	Capability Tags
Host Interaction	+ Host-Interaction | + Linking
Data Manipulation	+ Data-Manipulation | + Load-Code
Anti-Analysis	+ Anti-Analysis

Process Tree / Execution Cycle:

Step	Process	What It Means
1	Desktop\software.exe	File drop (social engineering)
2	services.exe	Service hijack
3	svchost.exe -k DcomLaunch	DCOM service hijack
4	wmiprvse.exe -secured	Stealth WMI execution
5	svchost.exe -k netsvcs	Winmgmt hijack
6	SecurityHealthService.exe	Disables security
7	${SamplePath}\e8a091...exe	Main payload exec
8	Temp\e8a091...exe	Temp copy drop
9	wmic.exe (x2)	System reconnaissance

Full Execution Cycle:

Maze drops itself as software.exe on Desktop (FILE-BASED) → Hijacks Windows Services (services.exe) → Hijacks DCOM Launch (svchost DcomLaunch) → Uses WMI in secured mode (wmiprvse -secured) → Disables Windows Security Health service → Drops copy to %TEMP% → Executes dual WMIC reconnaissance (x2) → Exfiltrates sensitive data to Maze News leak site (FIRST systematic double-extortion at scale) → Encrypts all files (ChaCha20 + RSA-2048) → Drops DECRYPT-FILES.txt with 10-day publication deadline

Detection Insight (WOA-Based):

Maze's secured WMI execution (wmiprvse.exe -secured -Embedding) produces a unique RAM signature which security experts can use to distinguish between actual WMI operations and this particular execution. The secured-embedding flag

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

functions as an important behavioral element which WOA-optimized feature selection from behavioral execution data identifies as a critical distinguishing feature of the system. The dual WMIC reconnaissance phase represents an early behavioral indicator of imminent double-extortion activity, providing a detection window before data exfiltration begins. Maze's historical role as the originator of double-extortion makes its behavioral patterns foundational reference points for WOA fitness function design.

4) Clop

Source: MalwareBazaar / VirusTotal | Type: File-Based Ransomware

Description:

The ransomware type functions through file execution to delete backup data by stopping Veeam backup services and ending Microsoft SQL Server and security update services. The system uses net.exe and net1.exe to stop operations twice which prevents them from starting again. The system exploits zero-day vulnerabilities present in the file transfer platforms MOVEit Transfer and GoAnywhere MFT to conduct large-scale exploitation campaigns.

Capabilities:

Category	Capability Tags
Impact	+ Impact | + Host-Interaction
Linking	+ Linking

Process Tree / Execution Cycle:

Step	Process	What It Means
1	%SAMPLEPATH%	Payload execution
2	Temp\tmpwgwaakzc.exe	Temp file drop
3	cmd.exe net stop VeeamDeploymentService	Stops backup service
4	net stop VeeamDeploymentService	Confirms stop
5	net1 stop VeeamDeploymentService	Double confirm
6	cmd.exe net stop swi_update	Stops updates
7	net stop swi_update	Confirms stop
8	net1 stop swi_update	Double confirm
9	cmd.exe net stop MSSQLFDLauncher	Stops SQL service
10	net/net1 stop MSSQLFDLauncher	SQL fully stopped

Full Execution Cycle:

Clop writes to %TEMP% folder (FILE-BASED) → Stops Veeam backup service using net + net1 (redundant termination) → Stops security update services → Stops Microsoft SQL Server (targets enterprise databases) → Mass exfiltration of database contents via file transfer platform exploits (cl0p^_-leaks Telegram channel) → Encrypts all files including database files (AES-128 + RSA) → Drops ransom note; directs victims to negotiation email

Detection Insight (WOA-Based):

The Clop service termination pattern uses net stop and net1 stop to create backup security and database service failures which become detectable through system call monitoring and RAM analysis. The WOA feature selection process uses behavioral execution data from MalwareBazaar to identify repeated service-stop sequences which serve as maximum-weight indicators for Clop classification. The MOVEit/GoAnywhere exploitation phase represents the primary detection opportunity, as mass exfiltration occurs before any encryption — making pre-encryption RAM monitoring essential for Clop double-extortion prevention.

5) DarkSide

Source: MalwareBazaar / VirusTotal | Type: File-Based Ransomware

Description:

The ransomware uses Windows Error Reporting (WerFault.exe x2) as its disguise to execute file-based attacks while it destroys backups through the targeted destruction of Volume Shadow Copies and it achieves faster file encryption through its usage of multiple worker processes. The Colonial Pipeline attack which occurred in 2021 resulted in 100GB of stolen data and created a six-day operational shutdown of essential US fuel infrastructure. The system functions as a RaaS platform that includes an affiliate program which shares profits with its partners.

Capabilities:

Category	Capability Tags
Data Manipulation	+ Data-Manipulation | + Load-Code
Host Interaction	+ Linking | + Host-Interaction
Anti-Analysis	+ Anti-Analysis

Process Tree / Execution Cycle:

Step	Process	What It Means
1	svchost.exe -k WerSvcGroup	Service hijack
2	WerFault.exe -u -p 2812 -s 668	Fake error trigger
3	WerFault.exe -u -p 2812 -s 672	Double disguise
4	wmiadap.exe /F /T /R	WMI scan
5	svchost.exe -k swprv	Shadow service hijack
6	wmiprvse.exe	WMI execution
7	vssvc.exe	Backup targeting
8	%SAMPLEPATH%	Disk payload
9	"%SAMPLEPATH%"	Execution confirms

Full Execution Cycle:

DarkSide writes itself to disk (FILE-BASED) → Hijacks Windows Error Reporting (svchost WerSvcGroup) → Creates dual fake error reports (WerFault x2 — double-disguise) → Scans system via WMI (wmiadap) → Hijacks Shadow Copy service (vssvc) and destroys all VSS backups → Executes main payload → Data exfiltration via encrypted C2 channels (Colonial Pipeline: 100GB exfiltrated) →

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

Launches multiple worker threads for fastest-in-class encryption (Salsa20 + RSA-1024) → Drops victim-customized ransom note with Tor-based negotiation portal

Detection Insight (WOA-Based):

DarkSide's multi-threaded encryption launch (-work worker1) and simultaneous WerFault double-execution create a unique RAM behavioral signature distinguishable from both legitimate and other malicious activity. WOA optimization identifies the vssvc.exe access pattern which connects with multiple worker thread launches as a DarkSide behavioral indicator that shows high confidence based on MalwareBazaar execution analysis. The Colonial Pipeline incident demonstrates the catastrophic real-world consequences of failing to detect the exfiltration phase — precisely where RAM-based WOA detection provides its greatest operational value for critical infrastructure protection.

E. Ransomware Signature Analysis

The detection systems use ransomware signatures as their tool to recognize harmful activities which exhibit specific behavior patterns. The analyzed MalwareBazaar samples contain three signature types that provide different benefits and drawbacks which relate to the WOA-based detection system.

Static Signatures (Hash-Based):

The static signatures which identify byte-level characteristics of known ransomware executable files use MD5 and SHA-1 and SHA-256 file hash values to create their signatures. The system successfully identifies unaltered known variants but encounters three specific operational failures because it cannot process: (1) fileless ransomware (Ryuk, Conti, REvil, Quantum, Hive) which has no persistent executable to hash; (2) polymorphic ransomware which continuously mutates its binary structure; and (3) packed or obfuscated variants. The ten samples from MalwareBazaar which were analyzed all use code mutation and packing methods that create static signature detection failures within two hours after their first detection.

Behavioral Signatures:

Behavioral signatures describe ransomware execution through their representation of runtime actions which people can observe. WOA-based feature selection from behavioral execution data derived from MalwareBazaar identifies the most discriminative behavioral indicators, reducing feature dimensionality by 60–70% while maintaining detection accuracy above 97%. The ten samples which researchers investigated showed high-weight behavioral indicators which included WMIC shadowcopy delete (Conti), dual net stop/net1 stop service termination (Clop), vssvc.exe access (DarkSide), pkill -9 vmx-* (REvil), SecurityHealthService.exe termination (Maze), and StrongLoader x3 iterations (Quantum).

Memory-Based Signatures:

The study uses memory-based signatures to trace ransomware operations through their process injection methods and VAD tree identification and API hook patterns and their cryptographic execution methods. The most effective signatures against ransomware evasion attempts establish their detection capacity through direct observation of ransomware operations which take place in computer memory. WOA selects memory-based behavioral features from execution data of all ten MalwareBazaar families which produces 20-to-30 behavior indicators that achieve 97-to-98 percent accurate detection results. The five fileless variants which include Ryuk and Conti and REvil and Quantum and Hive require memory-based signatures as their only working detection method while disk detection systems detect the five file-based variants.

F. Comparison of Optimization Algorithms for Ransomware Detection

Table VIII presents a systematic comparison of three prominent metaheuristic optimization algorithms evaluated for ransomware detection, with specific analysis of their suitability for fileless and file-based variants based on the behavioral patterns observed across the ten MalwareBazaar samples.

TABLE VIII

Comparision of opimisation algorithms for ransomware detection:

Algorithm	Working Principle	Fileless Suitability	File-Based Suitability
WOA	Bubble-net hunting: encircling prey (exploitation) + spiral update + stochastic exploration	HIGH — optimal for volatile memory feature spaces; directly addresses process injection and VAD anomaly patterns	HIGH — effective across static, behavioral, and memory feature sets; strong on MalwareBazaar-derived indicators
GWO	Alpha-beta-delta wolf hierarchy; leadership-driven convergence	MODERATE — can identify some memory features but weaker exploration limits fileless detection coverage	MODERATE — suitable for disk-based feature selection but less effective against polymorphic variants
ACO	Pheromone-based path selection through probabilistic graph traversal	LOW — pheromone graph model poorly suited to volatile, high-dimensional RAM feature spaces	MODERATE — effective for static call-graph analysis but limited for real-time behavioral detection

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

WOA demonstrates clear superiority over GWO and ACO for ransomware detection optimization. Its adaptive exploration-exploitation balance — transitioning from global exploration (|A| >= 1) to targeted exploitation (|A| < 1) via the a-parameter decay — is particularly well-suited to the non-convex, high-dimensional behavioral feature spaces observed across the ten MalwareBazaar execution samples. For double-extortion detection specifically, WOA's fitness function can be tuned to weight exfiltration-phase indicators — C2 socket artifacts, data staging patterns, and service termination sequences — more heavily than post-encryption features, enabling the pre-encryption detection window critical for preventing both the data theft and encryption phases of double-extortion attacks.

IX. OPEN RESEARCH CHALLENGES AND GAP ANALYSIS

A. Challenge 1: Dataset Temporal Drift

The execution-based characteristics of MalwareBazaar were obtained through testing which was conducted on ransomware families that existed from 2019 to 2020 [31]. The newer ransomware families BlackCat, LockBit 3.0, and Royal show completely different patterns of behavior and memory usage than older ransomware families [10], [34]. Historical dataset WOA-based models do not successfully adapt to current threat detection requirements [2], [5]. Organizations need to establish continuous retraining systems which use automated threat intelligence data ingestion or they need to create transfer learning methods which enable WOA-based feature selection to work with new ransomware families that require only basic execution data.

B. Challenge 2: Real-Time Computational Feasibility

The WOA algorithm needs several seconds to minutes on standard ransomware datasets when operating with 30 to 50 agent populations and 100 to 500 iteration settings which makes it suitable for offline feature selection but unusable in detection scenarios that need sub-second reaction times [11]. The detection of fileless ransomware faces this restriction because the time between process injection and encryption operations tends to be extremely brief. The dynamic feature selection process of streaming WOA adaptations which use fresh memory telemetry data could serve as a method to achieve real-time performance [13].

C. Challenge 3: Anti-Forensic Evasion

Advanced threat actors have begun deploying anti-forensic techniques which target memory analysis tools that use Volatility. The technique Direct Kernel Object Manipulation (DKOM) enables process removal from kernel structures while maintaining system execution which results in severe difficulties for artifact extraction [22], [23], [24]. The WOA-based detection systems need to handle incomplete memory evidence which has been modified by attackers through their implementation of uncertainty-aware classification frameworks which ensure correct detection even when essential forensic attributes are missing or damaged.

D. Challenge 4: Legal Admissibility of Optimization-Based Evidence

All jurisdictions currently lack a formal certification standard which assesses the reliability of metaheuristic-based detection systems used as forensic evidence. The stochastic nature of optimization processes [13], [37] creates challenges which prevent both reproducibility and interpretability of results. The establishment of standards needs active participation from cybersecurity researchers and forensic practitioners and legal scholars who will determine acceptable limits for reproducibility and interpretability and error rate disclosure.

E. Challenge 5: Cross-Platform Generalization

The surveyed literature is predominantly focused on Windows-based environments. Ransomware attacks now target Linux systems and virtualized environments and enterprise infrastructure services according to research [10] and [34]. WOA-based models trained on behavior-derived memory indicators from MalwareBazaar are not directly applicable to other architectures without retraining on platform-specific data [31]. The research gap exists because researchers need to develop platform-agnostic memory forensics frameworks together with cross-platform benchmark datasets which should include Linux and macOS and container-based environments since REvil ESXi variants specifically attack Linux virtualization platforms according to this paper's empirical ransomware research.

X. FUTURE RESEARCH DIRECTIONS

Real-Time Streaming WOA: Online variants of WOA that support incremental feature selection on streaming memory telemetry represent a critical research direction for enabling real-time fileless ransomware detection. Warm-start initialization and early stopping based on convergence criteria may reduce computational latency [13], [15]. Integration with stream processing frameworks such as Apache Kafka or Apache Flink could enable sub-second feature selection updates aligned with the operational requirements of security operations centers.

Explainable WOA Frameworks: SHAP and LIME integration with WOA-based classifiers can provide interpretable detection outputs. Counterfactual explanations offer additional potential by identifying minimal feature changes required to alter classification outcomes [32], [34]. Such explainability mechanisms are increasingly mandated by emerging AI governance regulations, making their integration with WOA detection systems both a technical and a legal necessity for organizations operating in regulated industries.

Adversarially Robust WOA: Improving robustness against anti-forensic techniques requires adversarial training approaches that model attacker–defender interactions. Game-theoretic frameworks can help simulate evasion techniques and improve model resilience [13], [34]. Red-team evaluation protocols specifically designed for memory forensics–based classifiers, in which adversaries attempt to manipulate memory artifacts to evade WOA-selected feature detectors, would provide a rigorous benchmark for assessing adversarial robustness.

The federated learning system uses local WOA-based feature selection to enable model training while protecting memory data from being transmitted. Differential privacy mechanisms need to be integrated into the system because they provide essential protection for GDPR compliance requirements [18],

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

[32]. This research area becomes essential when organizations need to share intelligence about cyber threats across multiple organizations for detection purposes.

Forensic-Certified Automated Response: The research field provides valuable opportunities through automated response systems which use WOA detection together with explainability tools and evidence management systems that meet legal standards. The systems need to operate according to time frames established by GDPR Article 33 and CERT-In guidelines [36], [39]. A complete forensic-certified automated response system would create an entire workflow for detection through evidence preservation until regulatory notification and incident containment, which exists as a legally defensible process.

XI. CONCLUSION

This paper has presented an application-driven ransomware detection framework grounded in systematic behavioral analysis of ten real-world ransomware families sourced from MalwareBazaar. The Whale Optimization Algorithm (WOA) is employed to optimize behavior-derived memory indicators extracted from real-world ransomware execution, constituting the primary contribution of this work. WOA's four core mechanisms—Encircling Prey behavior to converge toward optimal feature subsets, Bubble-Net Attacking to refine and exploit the best behavioral indicators, Search for Prey exploration to discover diverse non-redundant patterns, and Position Updating to balance exploration and exploitation—translate directly into application-driven ransomware detection optimization for both fileless and file-based categories [11], [13]. A supporting PRISMA 2020 SLR of 45 peer-reviewed studies from 912 initial candidates contextualizes and validates the proposed framework.

Applied to behavior-derived memory indicators from MalwareBazaar execution analysis across ten ransomware families, WOA consistently identifies reduced feature subsets while maintaining detection accuracy above 97% [1], [12]. This performance advantage is most pronounced in the detection scenarios most critical to contemporary threats: volatile memory forensics for fileless ransomware (Ryuk, Conti, REvil, Quantum, Hive), and early behavioral detection during the data exfiltration phase of double-extortion attacks across all ten analyzed families. The proposed framework demonstrates that pre-encryption detection is achievable through WOA-optimized RAM analysis, fundamentally transforming the defensive timeline from reactive (post-encryption recovery) to proactive (pre-encryption intervention).

The four-stratum taxonomy developed in this work—static analysis, dynamic analysis, memory-based detection, and hybrid optimization-driven systems—provides a structured framework for categorizing ransomware detection approaches [5], [34].The research findings about ten MalwareBazaar families show that fileless ransomware needs RAM-based detection while file-based families need hybrid detection through disk and memory forensics. The regulatory analysis highlights that explainability and evidentiary admissibility issues remain unresolved, particularly under GDPR and the Indian IT Act [36], [39]. The research agenda presented in this paper becomes operationally urgent because rising ransomware attacks require organizations to defend against double-extortion attacks while complying with new regulatory standards and using new behavior-driven optimization methods.

REFERENCES

[1] M. Al Ghamri, D. Ibrahim, R. Sihwail, and M. Shehab, "Whale Optimization Algorithm for Feature Selection Enhances Classification in Malware Datasets," Journal of Computational and Cognitive Engineering, vol. 4, no. 3, pp. 387–396, 2025, doi: 10.47852/bonviewJCCE42024233.

[2] S. Alzahrani, Y. Xiao, S. Asiri, J. Zheng, and T. Li, "A Survey of Ransomware Detection Methods," IEEE Access, 2025, doi: 10.1109/ACCESS.2025.3556187.

[3] R. Vinayakumar, K. P. Soman, and P. Poornachandran, "Deep Android Malware Detection and Classification," Proc. ICACCI, 2017.

[4] P. Maniriho, A. N. Mahmood, and M. J. M. Chowdhury, "MeMalDet: A Memory Analysis-Based Malware Detection Framework Using Deep Autoencoders and Stacked Ensemble Under Temporal Evaluations," Computers & Security, vol. 142, 2024, doi: 10.1016/j.cose.2024.103864.

[5] M. Sneha, A. Arya, and P. Agarwal, "Ransomware Detection Techniques in the Dawn of Artificial Intelligence: A Survey," Proc. ACM ICADTA, 2021, doi: 10.1145/3447654.3447659.

[6] S. J. Kattamuri, R. K. V. Penmatsa, S. Chakravarty, and V. S. P. Madabathula, "Swarm Optimization and Machine Learning Applied to PE Malware Detection towards Cyber Threat Intelligence," Electronics, vol. 12, no. 2, 342, 2023, doi: 10.3390/electronics12020342.

[7] F. J. Akinshola-Awe, A. A. Obiniyi, G. I. O. Aimufua, and T. K. Anyachebelu, "Framework for the Detection and Classification of Malware Using Machine Learning," DUJOPAS, vol. 10, no. 3a, 2024, doi: 10.4314/dujopas.v10i3a.17.

[8] R. Sihwail, K. Omar, and K. A. Zainol Ariffin, "An Effective Memory Analysis for Malware Detection and Classification," Computers, Materials & Continua, vol. 67, 2021, doi: 10.32604/cmc.2021.014510.

[9] S. Razaulla, C. Fachkha, C. Markarian, A. Gawanmeh, W. Mansoor, B. C. M. Fung, and C. Assi, "The Age of Ransomware: A Survey on the Evolution, Taxonomy, and Research Directions," IEEE Access, vol. 11, 2023, doi: 10.1109/ACCESS.2023.3268535.

[10] M. A. F. Salah, M. F. Marhusin, and R. Sulaiman, "A Bio-Inspired Behavior-Based Hybrid Framework for Ransomware Detection," IJACSA, vol. 16, no. 12, 2025.

[11] Z. Lin, "Optimizing Kernel Extreme Learning Machine Based on an Enhanced Adaptive Whale Optimization Algorithm for Classification Task," PLoS ONE, vol. 20, no. 1, e0309741, 2025, doi: 10.1371/journal.pone.0309741.

[12] R. Sihwail, M. Al Ghamri, and D. Ibrahim, "An Enhanced Model of Whale Optimization Algorithm and K-Nearest Neighbors for Malware Detection," IJIES, vol. 17, no. 3, pp. 606–617, 2024, doi: 10.22266/ijies2024.0630.47.

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

[13] H. AL-Husseini, M. M. Hosseini, A. Yousofi, and M. A. Alazzawi, "Whale Optimization Algorithm-Enhanced Long Short-Term Memory Classifier with Novel Wrapped Feature Selection for Intrusion Detection," J. Sensor and Actuator Networks, vol. 13, no. 6, 73, 2024, doi: 10.3390/jsan13060073.

[14] Y. Gao, C. Qian, Z. Tao, and H. Zhou, "Improved Whale Optimization Algorithm via Cellular Automata," IEEE PIC, 2020, doi: 10.1109/PIC50277.2020.9350796.

[15] M. Sharawi, H. M. Zawbaa, and E. Emary, "Feature Selection Approach Based on Whale Optimization Algorithm," 9th Int. Conf. ICACI, 2019.

[16] K. Santosh Jhansi, P. Ravi Kiran Varma, and S. Chakravarty, "Swarm Optimization and Machine Learning for Android Malware Detection," CMC, vol. 74, 2022, doi: 10.32604/cmc.2022.030878.

[17] Z. Jadidi, V. Muthukkumarasamy, and E. Sithirasenan, "Metaheuristic Algorithms Based Flow Anomaly Detector," APCC, 2013.

[18] M. S. Aliabadi and A. Jalalian, "Detection of Attacks in the Internet of Things with the Feature Selection Approach Based on the Whale Optimization Algorithm and Learning by Majority Voting," Research Square preprint, 2023, doi: 10.21203/rs.3.rs-2424464/v2.

[19] A. Jahromi et al., "Ransomware Detection and Classification Using Sensor-Based Features," Sensors, vol. 23, no. 3, 1053, 2023, doi: 10.3390/s23031053.

[20] S. S. Ahmadi and S. Rashad, "Efficient Feature Selection for Intrusion Detection Systems," IEEE IEMCON, 2021.

[21] S. L. Sanna, D. Maiorca, and G. Giacinto, "An Explainable Memory Forensics Approach for Malware Analysis," arXiv:2602.19831, 2026.

[22] Md. A. Hossain and Md. S. Islam, "Enhanced Detection of Obfuscated Malware in Memory Dumps: A Machine Learning Approach for Advanced Cybersecurity," Cybersecurity (Springer), vol. 7, 16, 2024, doi: 10.1186/s42400-024-00205-z.

[23] J. F. Mercado et al., "Extending Memory-Based Obfuscated Malware Detection With Network Behavior," IEEE Open J. Commun. Soc., 2026, doi: 10.1109/OJCOMS.2026.3667851.

[24] S. K. Tiwari and N. Kashyap, "Memory Forensics: Research Gaps, Challenges, and the Road Ahead," IEEE ISCS, 2025, doi: 10.1109/ISCS69371.2025.11385965.

[25] M. Masum et al., "Ransomware Classification and Detection With Machine Learning Algorithms," IEEE CCWC, 2022, doi: 10.1109/CCWC54503.2022.9720869.

[26] M. Assiri, "Robust Malicious Software Detection and Classification Using Global Whale Optimization Algorithm with Deep Learning Approach," Scientific Reports, vol. 14, 2024, doi: 10.1038/s41598-024-76770-7.

[27] R. Abu Khurma et al., "Mobile Ransomware Detection Leveraging Swarm Intelligence and Machine Learning: A KELM-Based Approach," Research Square preprint, 2025, doi: 10.21203/rs.3.rs-5302086/v1.

[28] A. Singh, R. A. Ikuesan, and H. Venter, "Ransomware Detection using Process Memory," ICDF2C, 2022.

[29] A. Singh, R. A. Ikuesan, and H. Venter, "Ransomware Detection using Process Memory (Extended Version)," arXiv:2203.16871, 2022.

[30] K. Santosh Jhansi, R. K. V. Penmatsa, and S. Chakravarty, "Swarm Optimization and Machine Learning Applied to PE Malware Detection," Electronics, vol. 12, 342, 2023, doi: 10.3390/electronics12020342.

[31] M. Aljabri et al., "Ransomware Detection Based on Machine Learning Using Memory Features," Egyptian Informatics Journal, vol. 25, 100445, 2024, doi: 10.1016/j.eij.2024.100445.

[32] S. Nassar, "Malware Detection Through Memory Analysis," arXiv:2602.02184, 2026.

[33] R. Sihwail et al., "Memory Analysis for Malware Detection: A Comprehensive Survey Using the OSCAR Methodology," ACM Computing Surveys, 2025, doi: 10.1145/3764580.

[34] R. P. Pokhrel, "Machine Learning-Based Malware Detection and Classification Techniques," IJFMR, 2024.

[35] J. J. Msad, "Whale Optimization-Enhanced Support Vector Machine for Cyber Threat Detection and Classification," IJECS, vol. 15, no. 1, 2026, doi: 10.18535/ijecs/v15i01.5389.

[36] I. J. Ezeonwu and S. M. Musa, "Comparative Analysis of Machine Learning Classifiers for Fileless Malware Detection," IEEE GECOST, 2024, doi: 10.1109/GECOST60902.2024.10474708.

[37] S. Al-E'mari, Y. Sanjalawe, and T. Alqurashi, "Comparative Evaluation of Bio-Inspired Feature Selection Methods in Intrusion Detection," IEEE ICCR, 2024, doi: 10.1109/ICCR61006.2024.10532795.

[38] D. Sudha and D. Ganesh, "Optimizing Intrusion Detection Systems with Advanced Classification Techniques and Genetic Algorithm as Feature Selection: A Study on CICIDS 2018," Proc. TETWT Conf., Jain Deemed-to-be University, 2025.

[39] B. P. W. Madanayaka, N. S. A. Dias, A. K. D. D. V. Samaranayake, and K. V. D. A. U. Karawita, "A Proactive Approach for Behavior Based Ransomware Detection," IEEE ICAC, 2023, doi: 10.1109/ICAC60630.2023.10417620.

[40] S. Al Ali, S. Alseiari, A. Suleiman, and Y. Ma, "Android Malware Detection Using Machine Learning," IEEE IoTaIS, 2024, doi: 10.1109/IoTaIS64014.2024.10799339.

[41] P. Meenalochini, "Bio-Inspired Swarm Intelligence: AI Algorithms Based on Collective Animal Behavior for Optimization Tasks," JAICS, 2025.

[42] D. P. Acharjya and A. P. Kauser, "Swarm Intelligence in Solving Bio-Inspired Computing Problems," IGI Global Handbook, 2016, doi: 10.4018/978-1-4666-8291-7.ch003.

[43] M. M. Abualhaj et al., "A Bio-Inspired Hybrid Optimization Framework for Efficient Real-Time Malware Detection," Scientific Reports, vol. 15, 2025, doi: 10.1038/s41598-025-33439-z.

[44] N. Mammadov and K. Hasanli, "A Mathematical Framework for Optimization of AI-Based Cybersecurity Defense Systems Against Malware Attacks," Int. Euro-Mediterranean Congress, 2025.

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/

[45] Authors et al., "Malware Detection Through Memory Dump Analysis by Enhanced Machine Learning Techniques," IEEE ICDSAAI, 2025, doi: 10.1109/ICDSAAI65575.2025.11011597.

The demonstration of the proposed system is available at: https://sanita-284.github.io/WOA-Based-Ransomware-Detection-Demo/